UDHAYA ADHITHYA VARMAN

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SUMMARY

Passionate Artificial Intelligence graduate with a strong foundation in machine learning and problem-solving. Eager to apply AI knowledge and technical skills to develop innovative solutions for real-world challenges and contribute to organizational success.

EXPERIENCE

Artificial Intelligence Engineer Jun 2026 – Present
Unizen Technologies *Bengaluru, India*

- Contributing to live projects in Artificial Intelligence, Machine Learning, and Embedded Systems.
- Applying technical skills to real-world projects while gaining hands-on industry experience and collaborating with experienced professionals.

EDUCATION

Sathyabama Institute of Science and Technology of Engineering Aug 2022 – Jul 2026
Bachelor of Computer Science Engineering (CSE) – AI
CGPA: 8.0

Siddharatha Junior College Mar 2022
Higher Secondary Education
CGPA: 6.57

Gangothri Vidyalaya School
Mar 2018
Secondary Education
CGPA: 9.3

PROJECTS

- **Domain-Specific AI Research Assistant with Knowledge Graph** Ollama · Hybrid RAG · ChromaDB · Streamlit | Github

  I developed a Domain-Specific AI Research Assistant with Knowledge Graph that ingests academic PDFs and answers research queries using a Hybrid RAG pipeline combining ChromaDB dense retrieval and BM25 sparse search fused via Reciprocal Rank Fusion. The system verifies every generated claim for hallucinations using an NLI cross-encoder model and auto-builds an interactive multi-paper knowledge graph to enable graph-based reasoning across entities like models, datasets, and metrics.

- **Multi-Agent AI Research Assistant** LLaMA 3.3-70B · Groq · ChromaDB · Semantic Scholar · FastAPI· Streamlit | Github
  I developed a Multi-Agent AI Research Assistant that orchestrates 5 specialized AI agents — Manager, Searcher, Summarizer, Critic, and Writer — to automate the academic research process end-to-end. The system searches over 200 million papers from Semantic Scholar using custom relevance scoring combining semantic similarity, citation impact, and recency, then produces structured 7-section literature reviews complete with gap analysis and citation validation.

- **SignBridge — Real-Time ISL Translator with Edge Deployment** MediaPipe · CNN-LSTM · ONNX · OpenCV | Github
  I developed a real-time Indian Sign Language alphabet recognition system that detects and translates all 26 ISL hand signs using a sliding 30-frame CNN-LSTM temporal model with multi-head attention. The system uses MediaPipe Tasks API for automatic single/two-hand landmark extraction, achieves 99.74% validation accuracy trained on CPU in 22 minutes, and deploys via ONNX FP32 at 353 FPS.

SKILLS

- **Languages & Tools:** Python, SQL, Streamlit.
- **AI & ML:** RAG Pipelines, Multi-Agent Systems (LangGraph), NLP, Deep Learning,
- **Cloud & MLOps:** MLflow, Docker, GitHub Actions,

CERTIFICATIONS

- Certification in Artificial Intelligence
- Certification in Deep Learning Fundamental
- Certification in Machine Learning
- Certification in Prompt Engineering for Everyone
- Certification in Python for Database